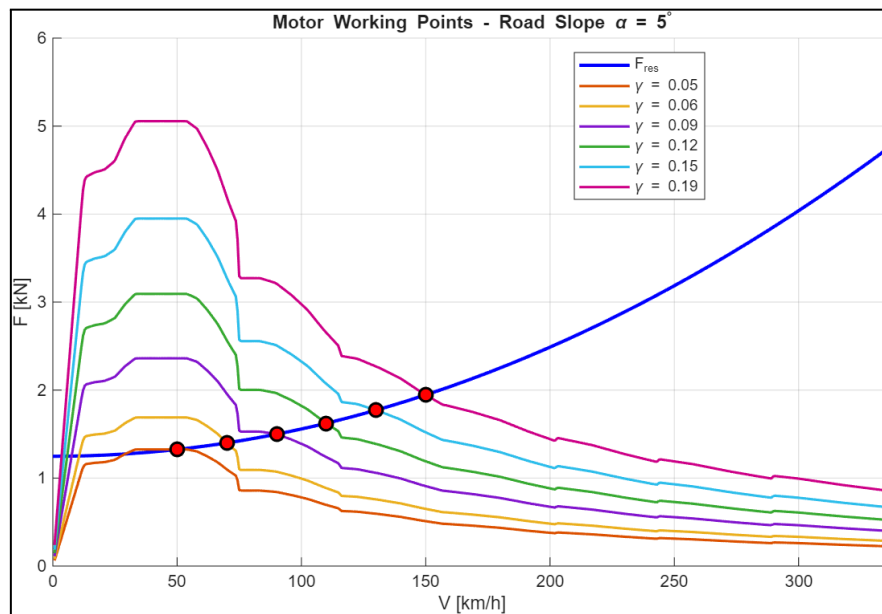
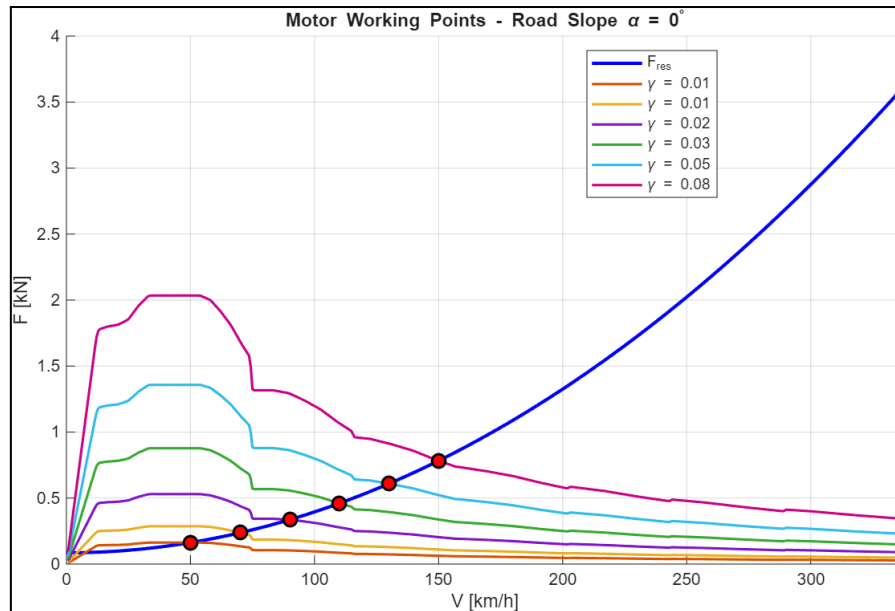


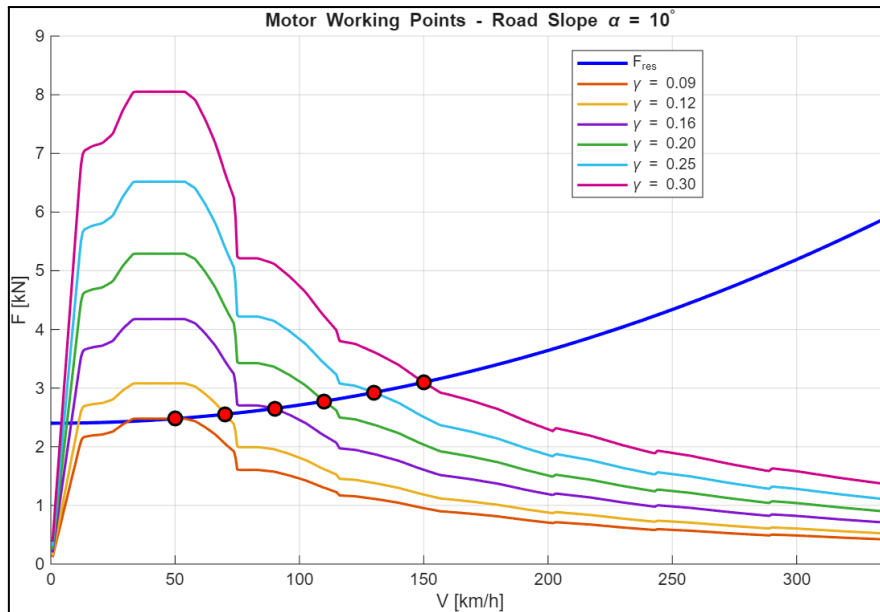
VEHICLE DYNAMICS AND CONTROL- Longitudinal Model

CAR SEGMENT - Sports Car

PART-1

a. Motor Operating Points





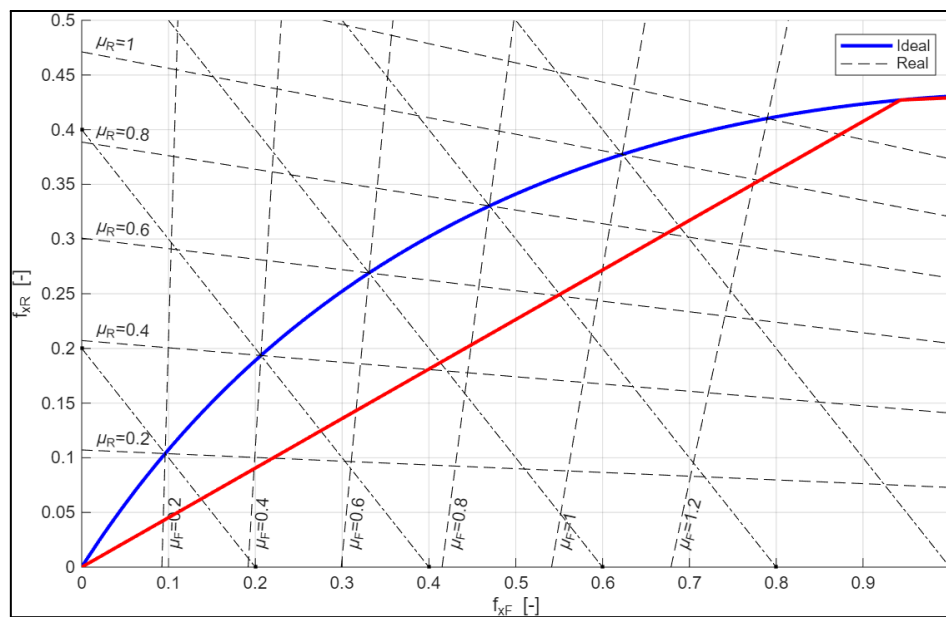
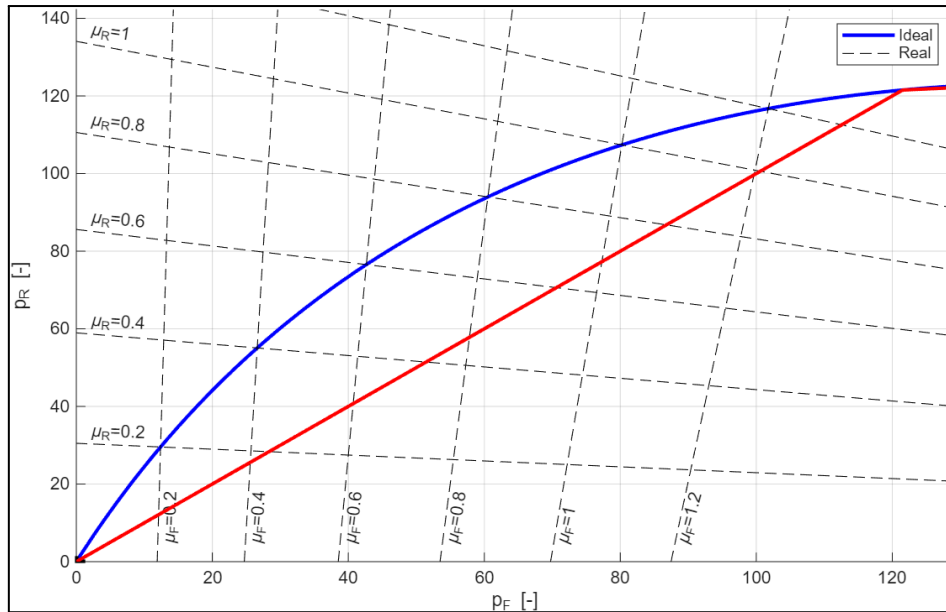
Motor Operating Points Tabular Result

Alpha_deg	V_kmh	Fx_N	gamma_percent
0	50	162.755203	0.612147493
	70	237.162611	1.076186264
	90	336.372487	1.992559057
	110	460.384833	3.300652525
	130	609.199648	5.107111147
	150	782.816932	7.651879789
5	50	1325.22787	4.984386986
	70	1399.63528	6.351204574
	90	1498.84515	8.878661589
	110	1622.8575	11.63480705
	130	1771.67231	14.85248301
	150	1945.2896	19.01481885
10	50	2478.20463	9.320910917
	70	2552.61204	11.58313278
	90	2651.82191	15.70851354
	110	2775.83426	19.90088226
	130	2924.64907	24.51824774
	150	3098.26636	30.28493734

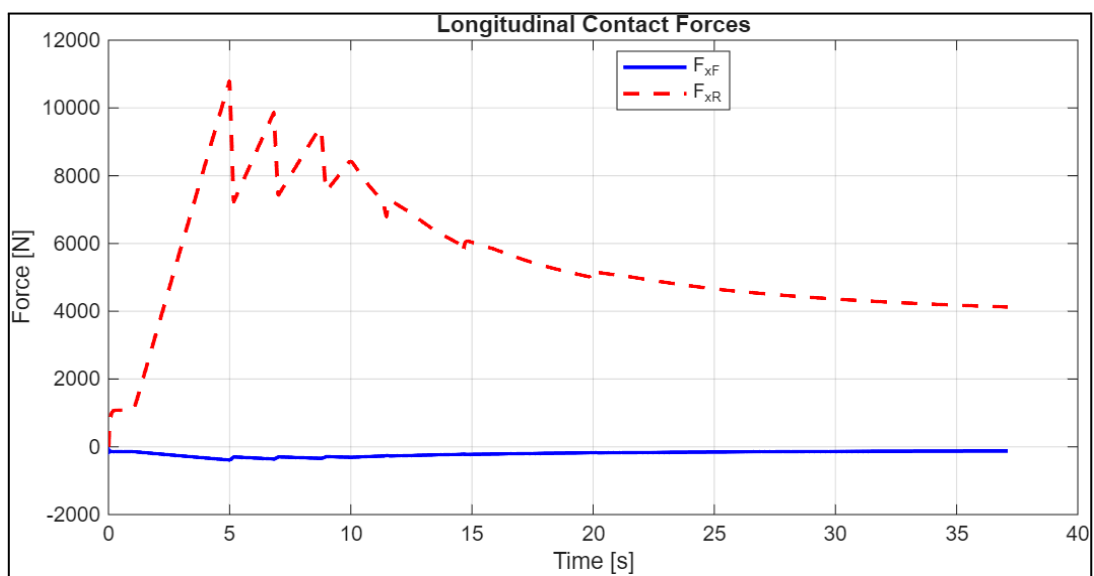
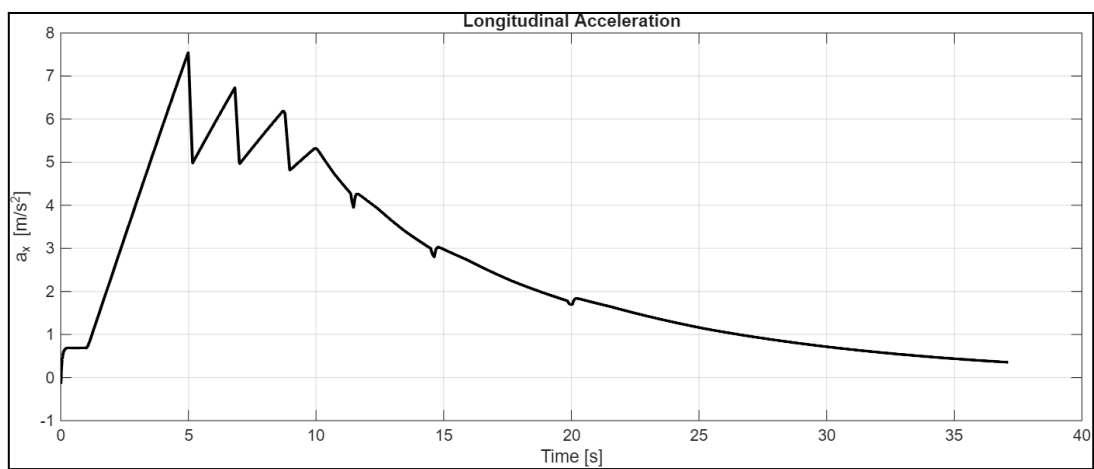
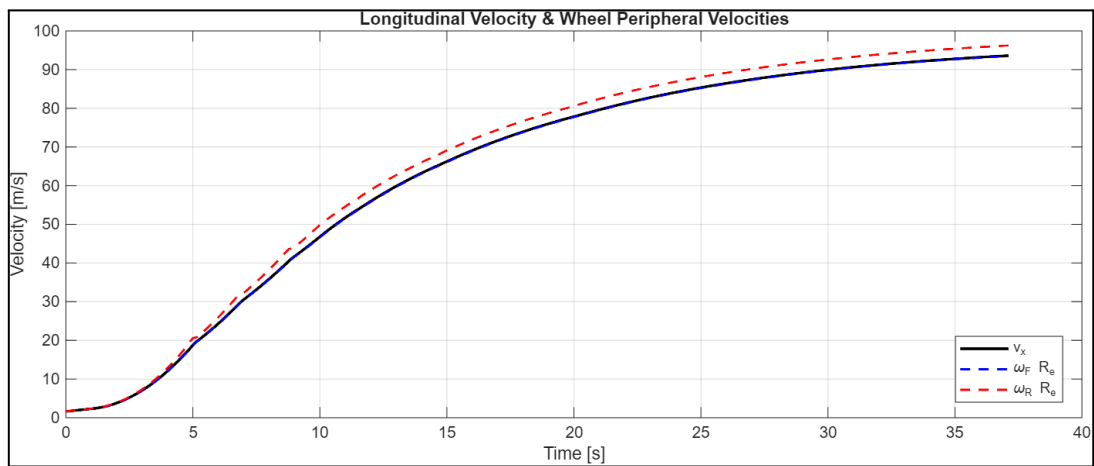
The motor operating points are marked on the plots.

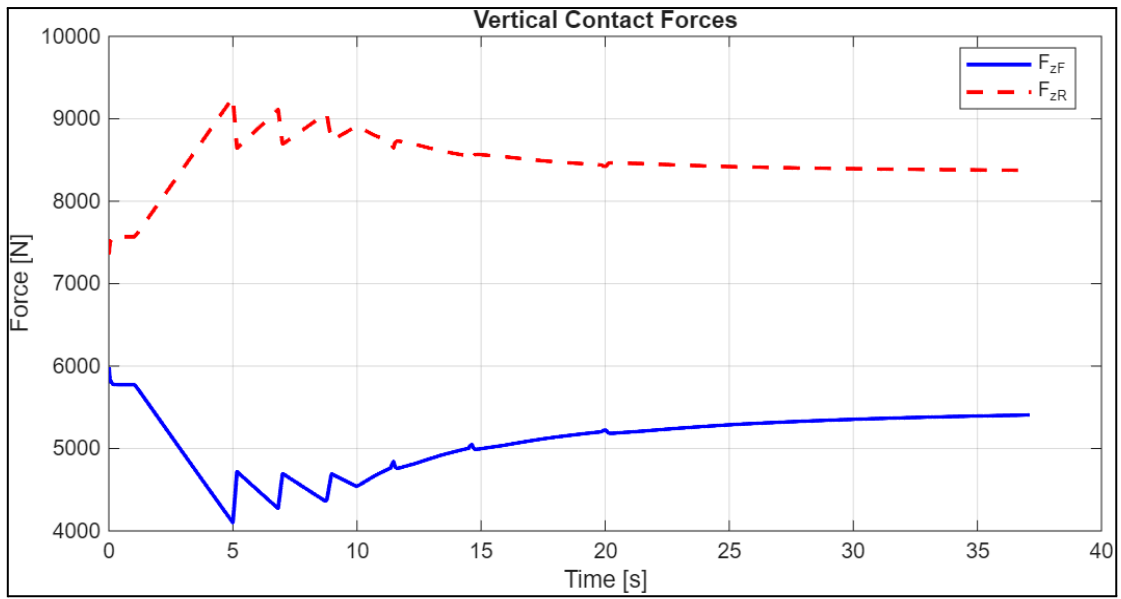
b. Brake Mechanical Distributor

From the graph obtained by plotting the ideal brake distribution, the p_{Dis} value is found to be almost equal to **121.5 bars**.

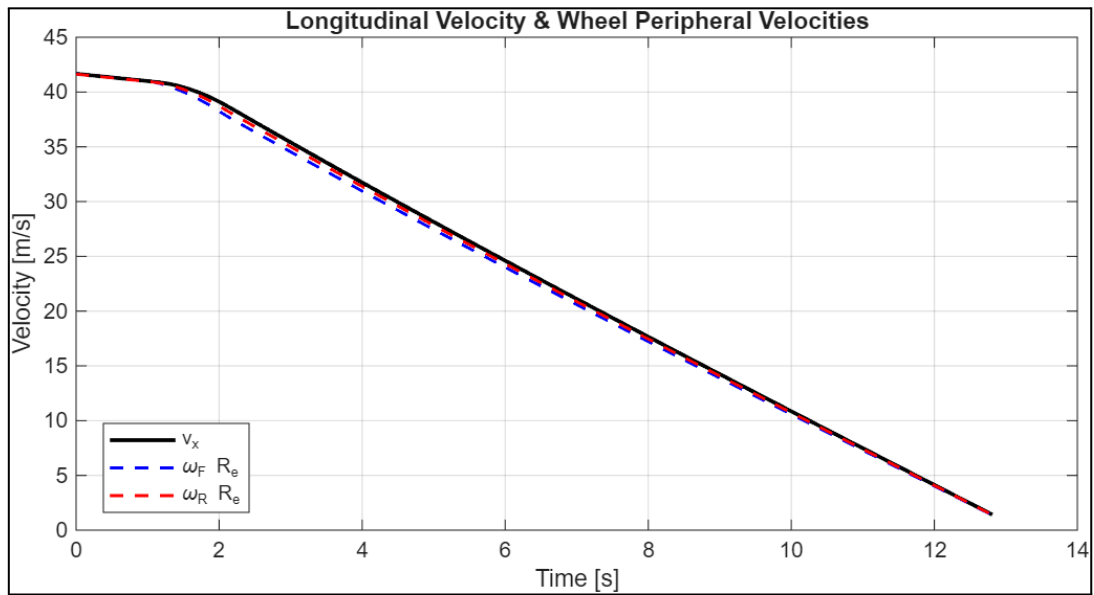


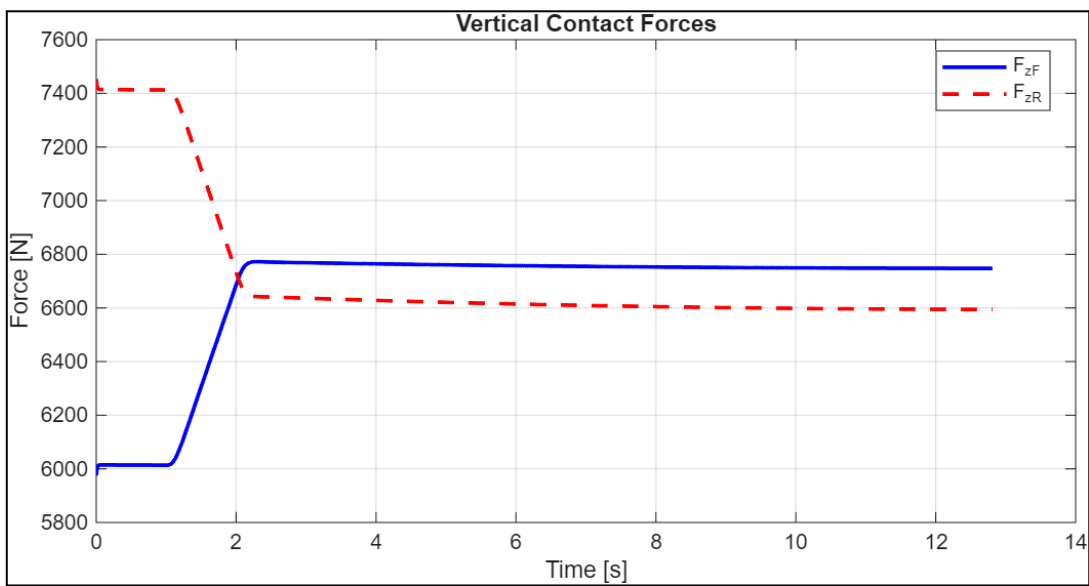
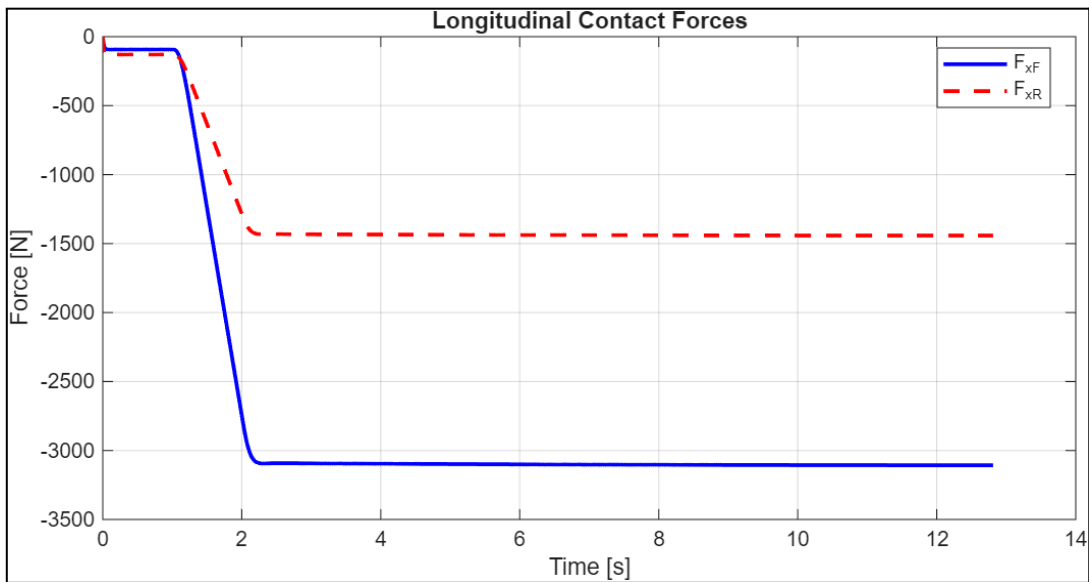
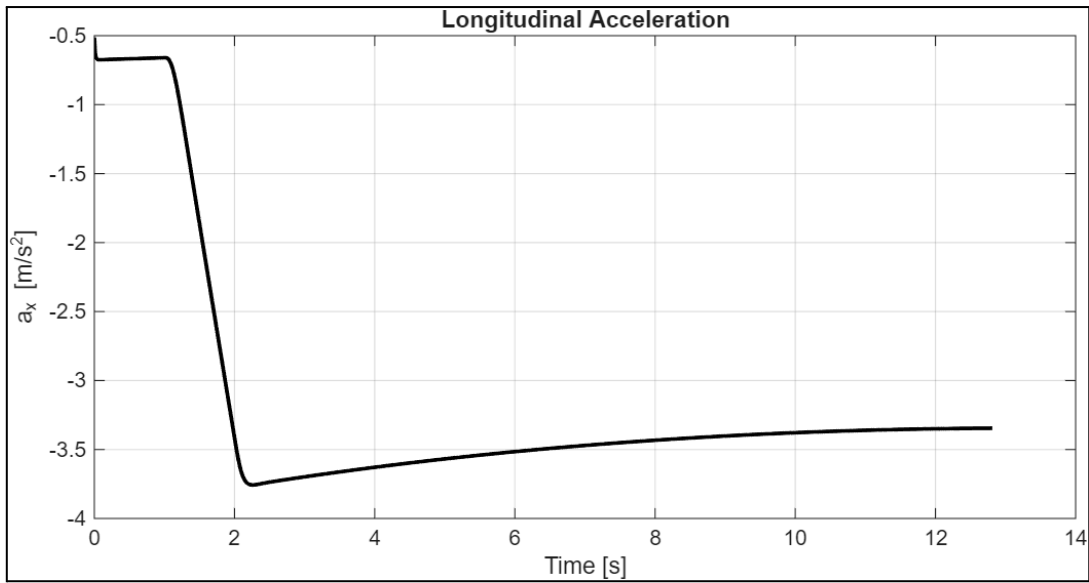
c. Straight Line Acceleration Maneuver





d. Straight Line Braking Maneuver

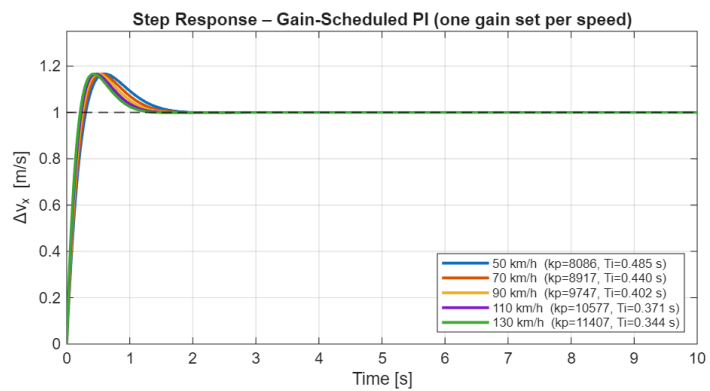
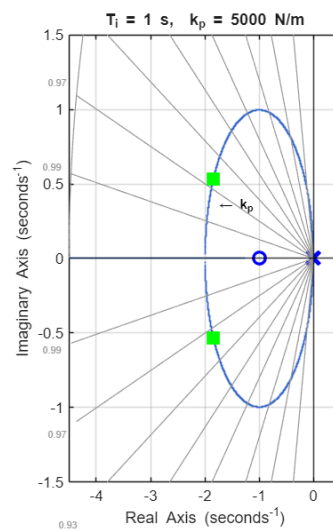
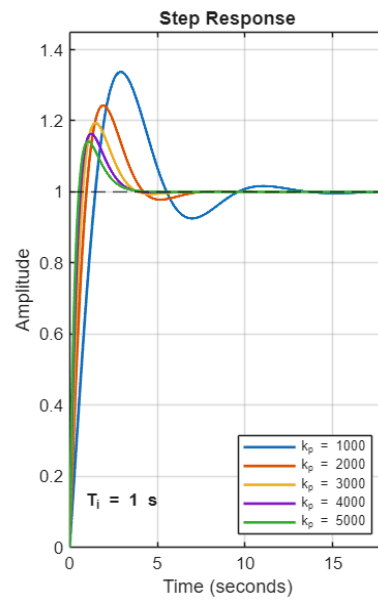




PART - 2

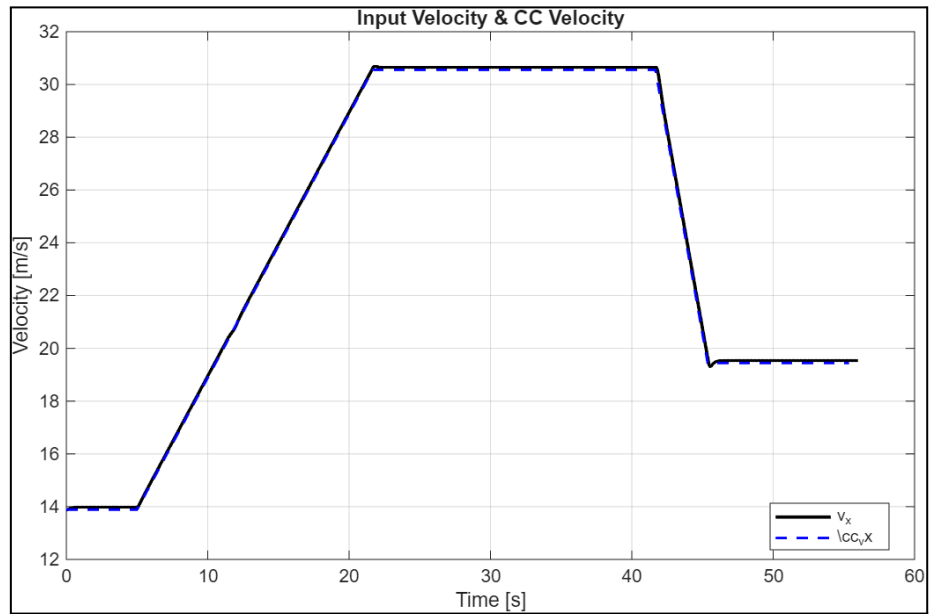
A. Cruise Control

- Effect of control gain K_p

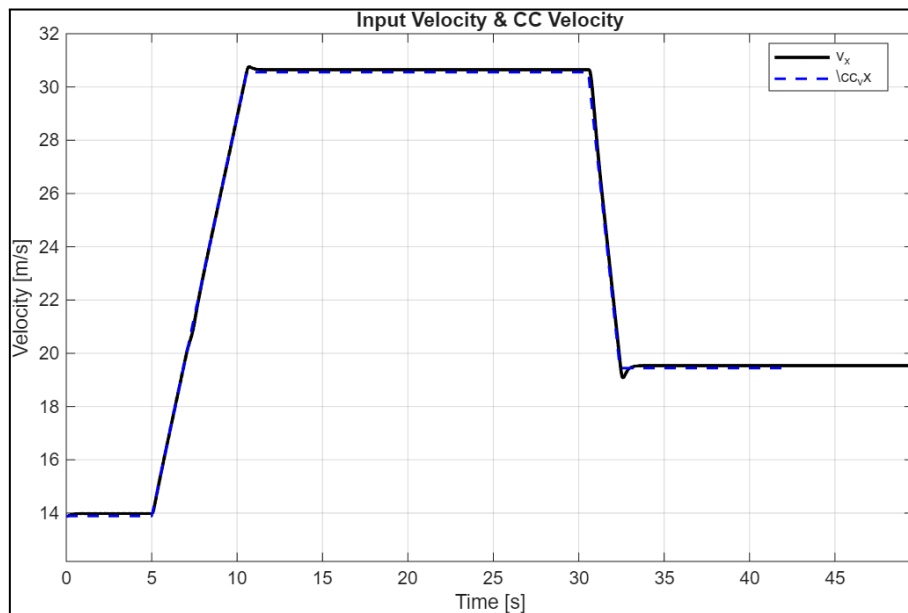


CRUISE CONTROL - PI GAIN TABLE (zeta_des = 0.85)				
v [km/h]	omega_n [rad/s]	kp [N/(m/s)]	Ti [s]	zeta_check
50	3.500	8086.42	0.4854	0.8500
70	3.860	8916.51	0.4400	0.8500
90	4.220	9746.59	0.4024	0.8500
110	4.580	10576.68	0.3707	0.8500
130	4.940	11406.77	0.3437	0.8500

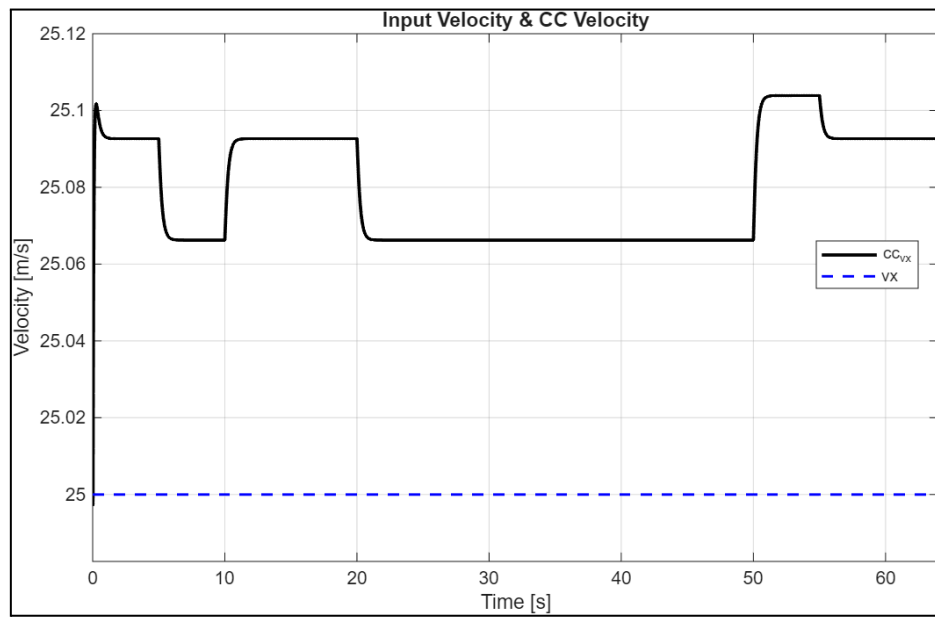
- For given speed reference, $a^+ = 1 \text{ m/s}^2$, $a^- = -3 \text{ m/s}^2$



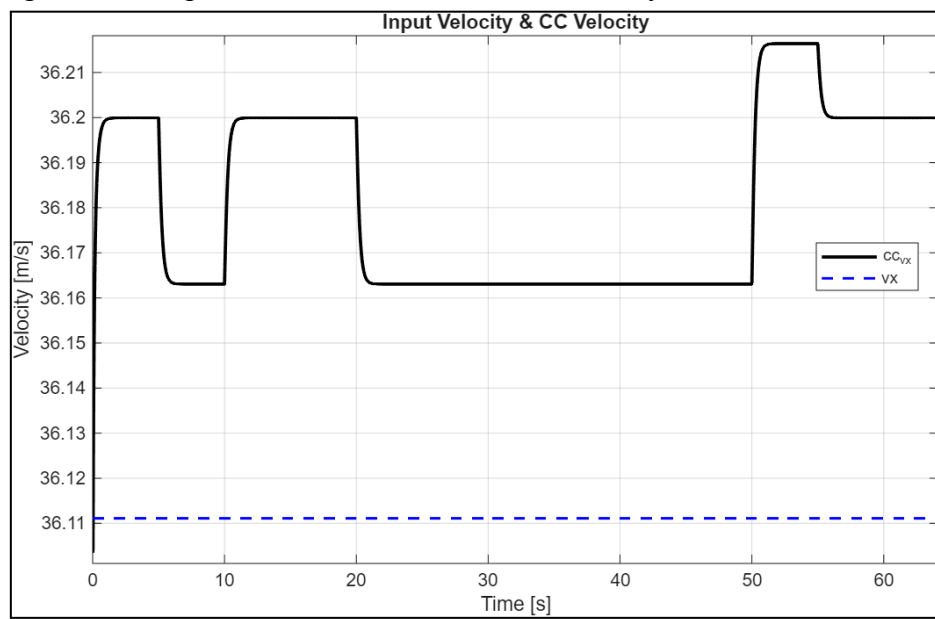
- For given speed reference, $a^+ = 3 \text{ m/s}^2$, $a^- = -6 \text{ m/s}^2$



- For given wind speed reference with constant velocity, 90 km/h



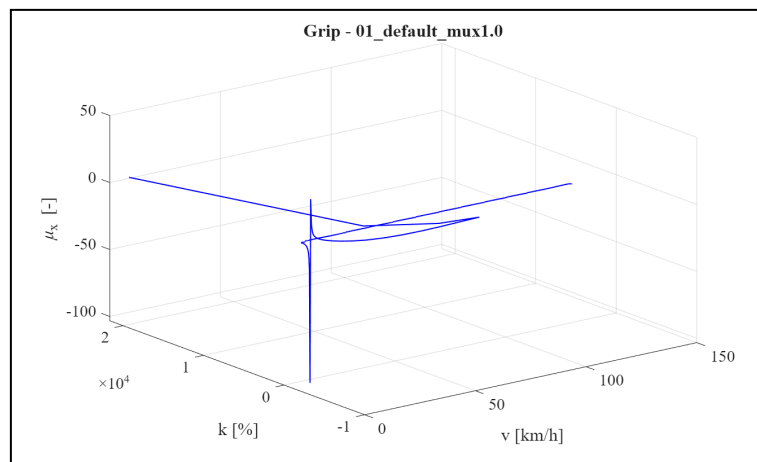
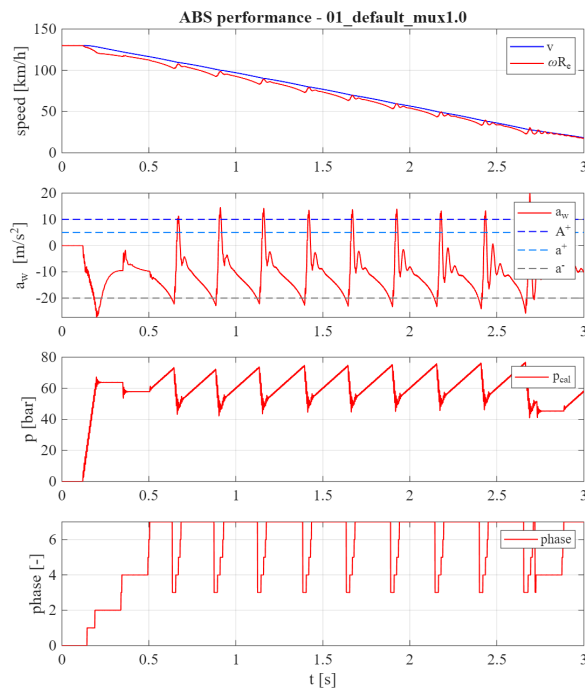
- For given wind speed reference with constant velocity, 130 km/h



B. ABS Tuning and Control

- Baseline ABS setup

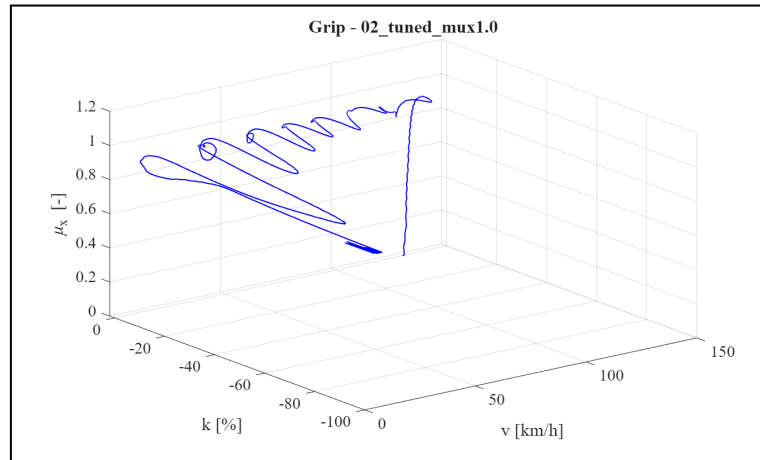
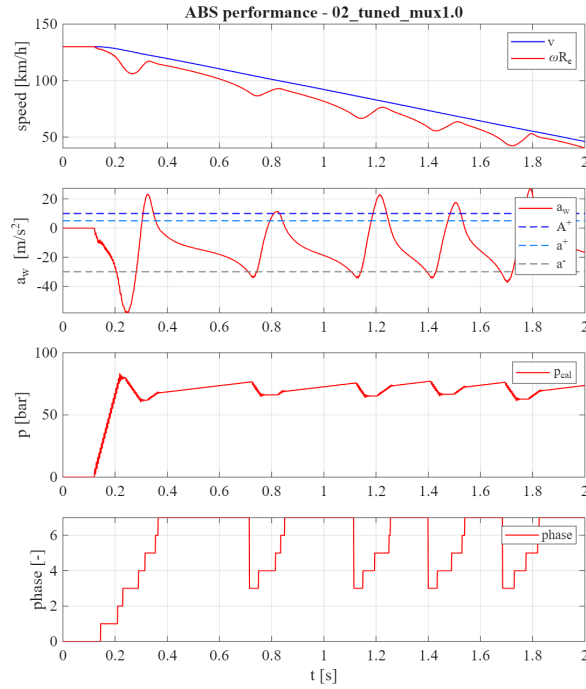
$am=-20$, $ap=5$, $Ap=10$, $dp3=1200$, $dp5=400$, $dp7=100$ bar/s



This is the untuned starting point provided with the model. Conservative deceleration thresholds keep the wheel slip well below the tire's peak-grip slip ($\sim 15\%$), giving a safe but long stopping distance.

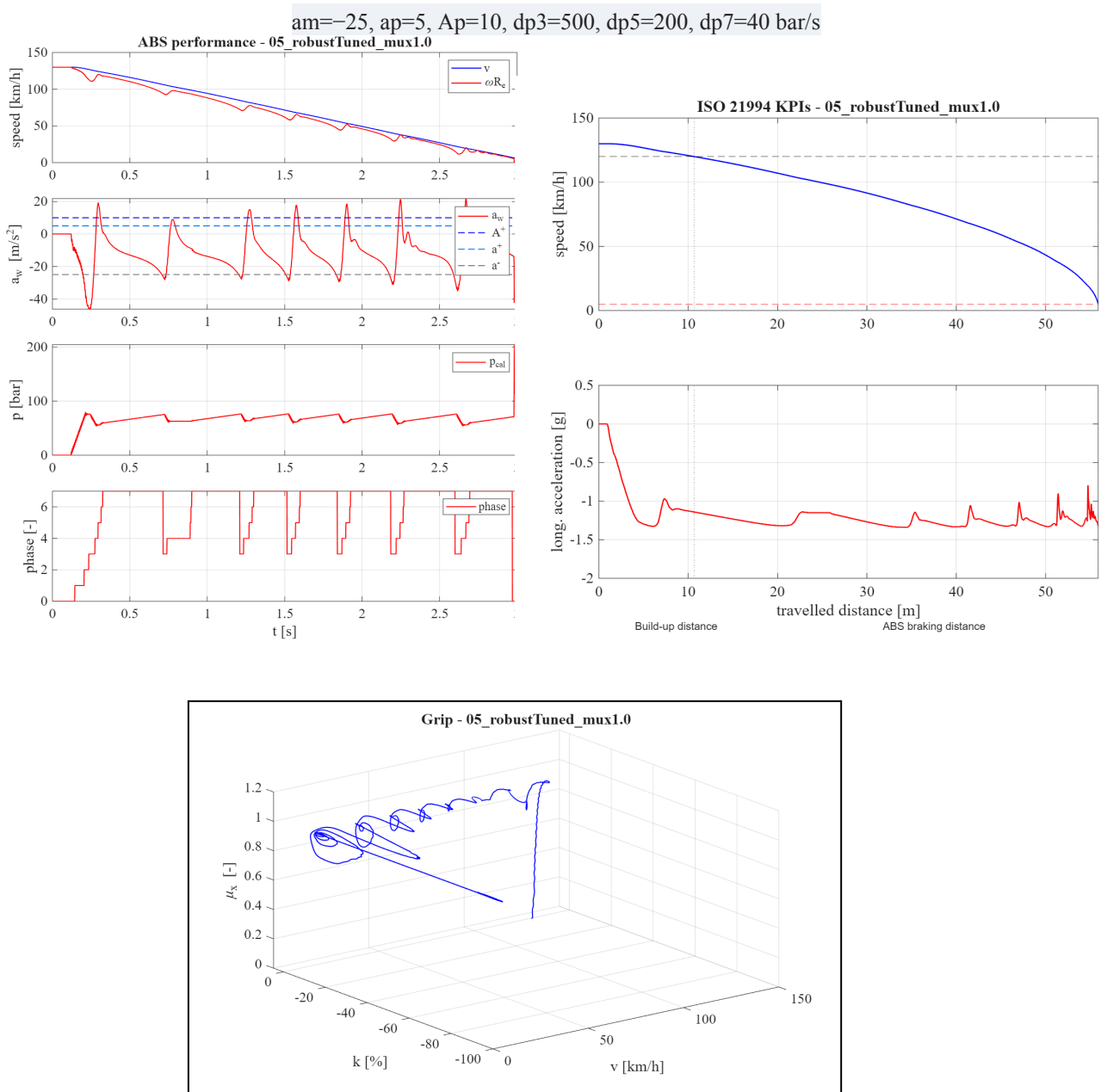
- Tuned Setup 1

am=-30, ap=5, Ap=10, dp3=300, dp5=150, dp7=25 bar/s



In this tuned setup, deceleration threshold a^- lowered ($-20 \rightarrow -30$) to let the wheel decelerate harder before triggering pressure release; all three pressure rates cut roughly 4 $\times$ ($1200/400/100 \rightarrow 300/150/25$) so the controller modulates pressure more gently and keeps the wheel operating closer to the tire's peak-grip slip region. Shortens stopping distance by ~ 7.6 m but pushes mean slip up and causes brief excursions above the 15% slip threshold.

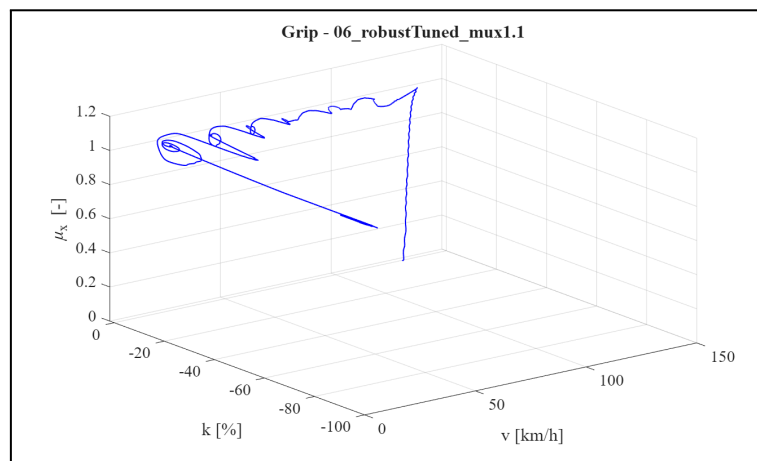
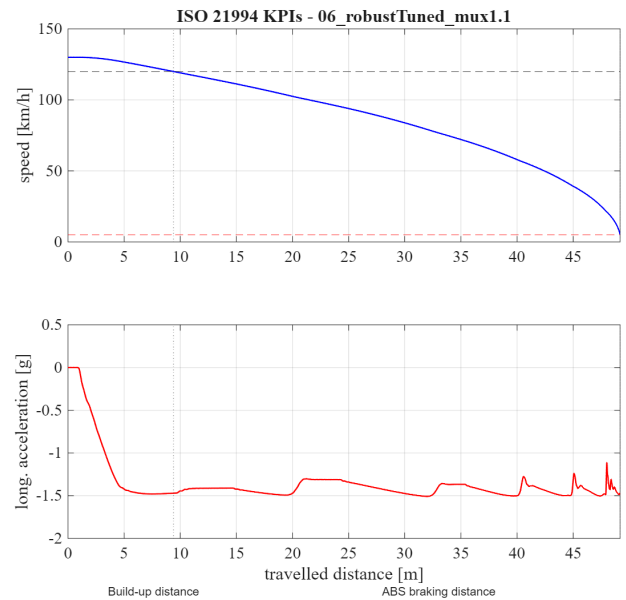
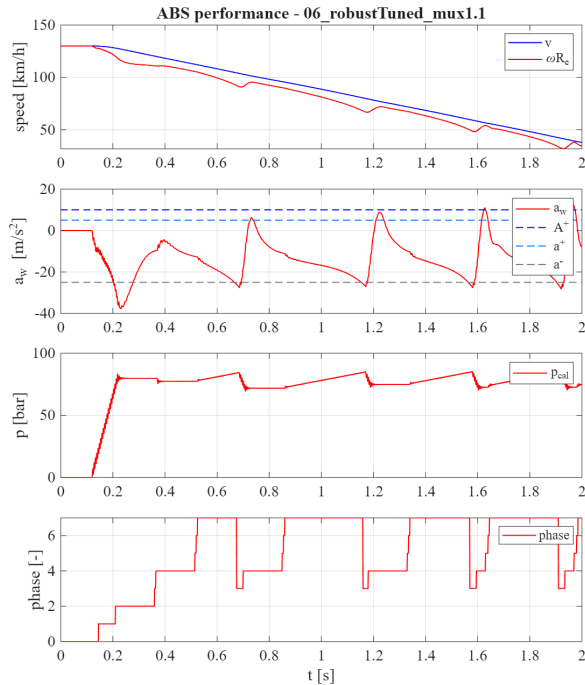
- Nominal Grip , $\mu_x = 1.0$



This setup serves as the reference point for the $\pm 10\%$ comparison simulations.

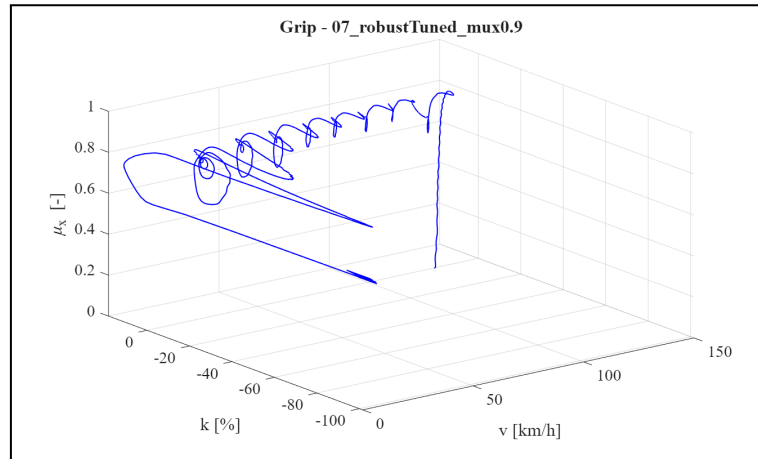
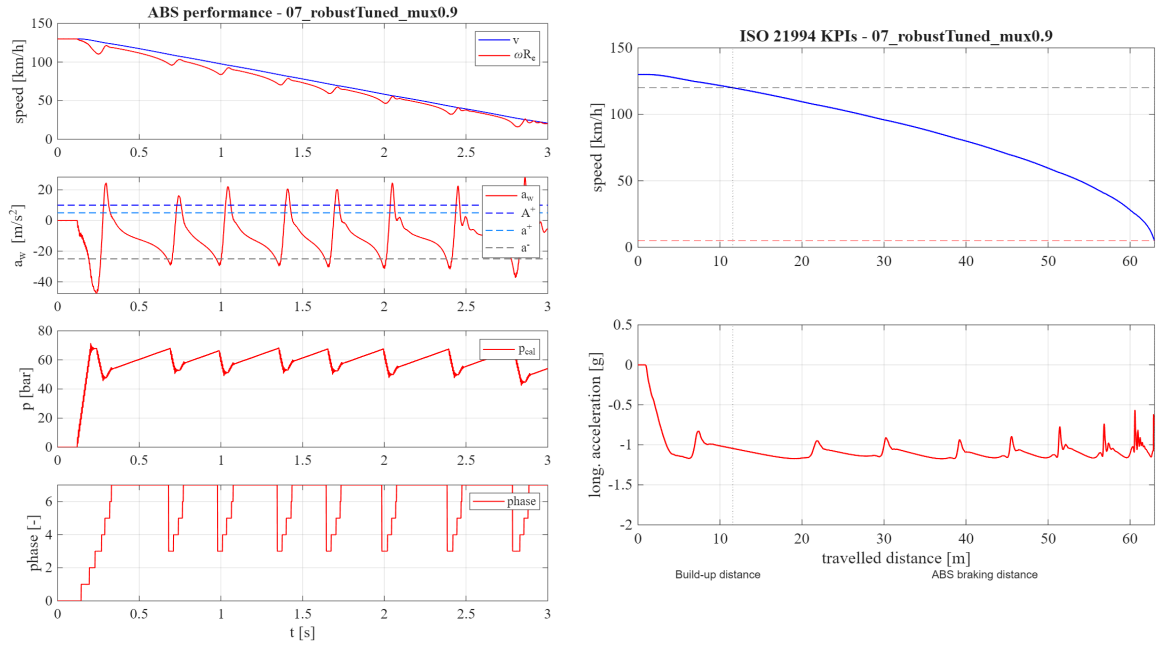
- Higher grip, $\mu_x = 1.1$ (+10%)

There is no change in the controller, only the tire's available grip increased by 10%. Stopping distance drops to 49.18 m with zero slip overshoot — more grip is simply "free" extra performance, the controller does not need adjustment.



- Lower grip, $\mu_x = 0.9$ (−10%)

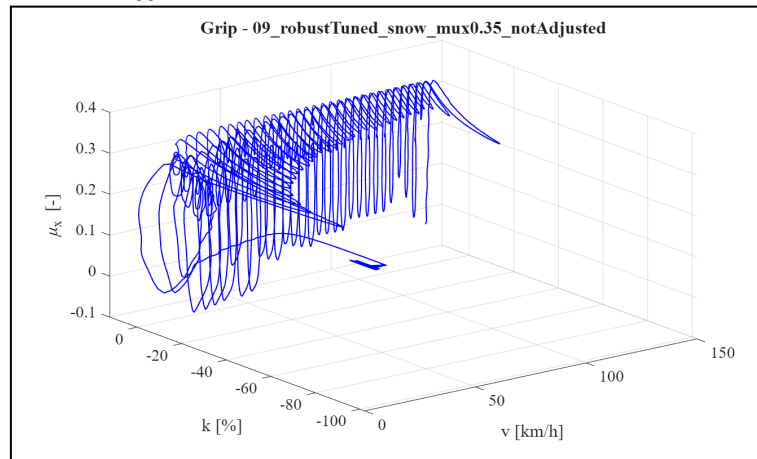
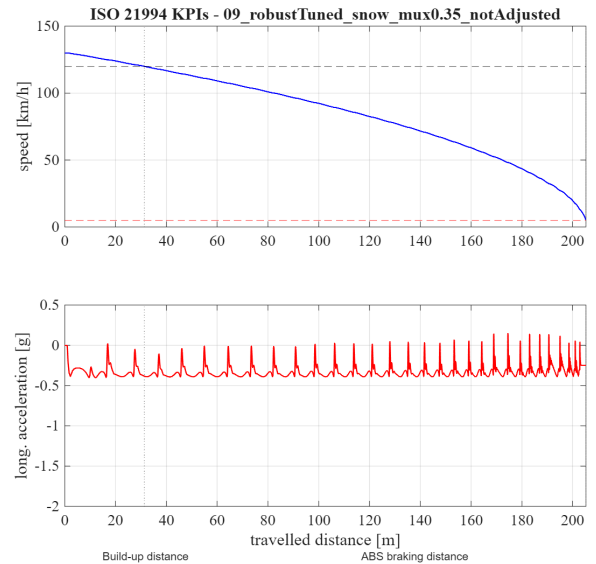
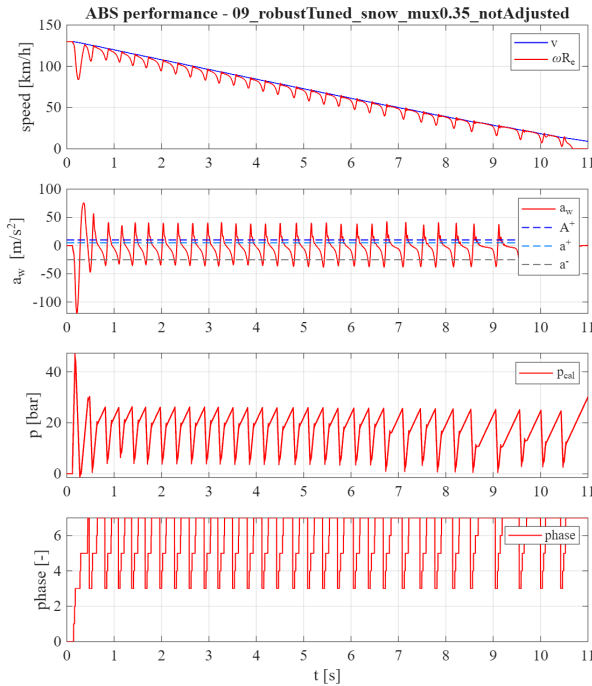
There is no change in the controller, only the tire's available grip reduced by 10%. Stopping distance grows to 62.98 m, still with zero slip overshoot — confirms the robust gain set (Case 3) tolerates a 10% grip drop without needing re-tuning, unlike the aggressively tuned gains from Case 2 (which gave 28% slip overshoot under the same grip drop).



- Snowy Conditions unadjusted $\mu_x = 0.35$

$a_m = -25$, $a_p = 5$, $A_p = 10$, $dp_3 = 500$, $dp_5 = 200$, $dp_7 = 40$ bar/s, $\mu_x = 0.35$

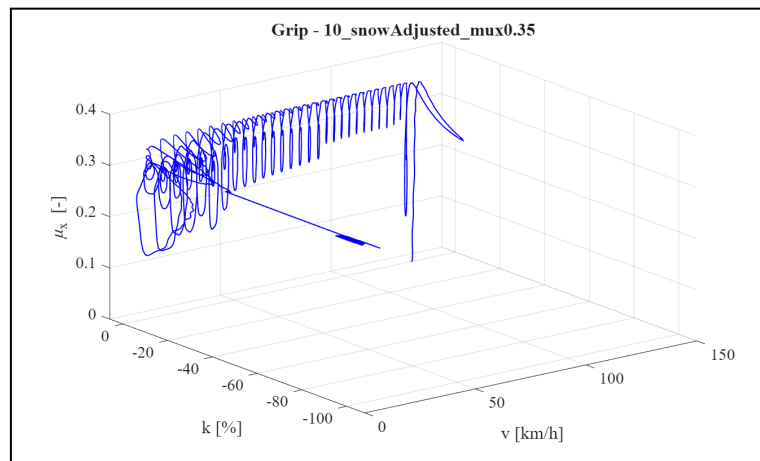
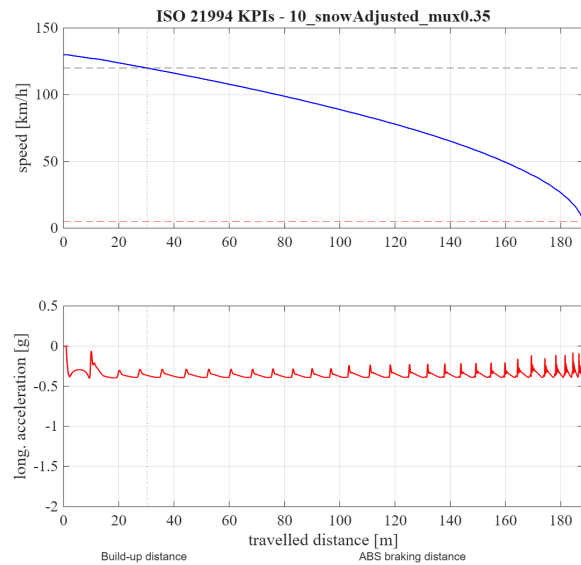
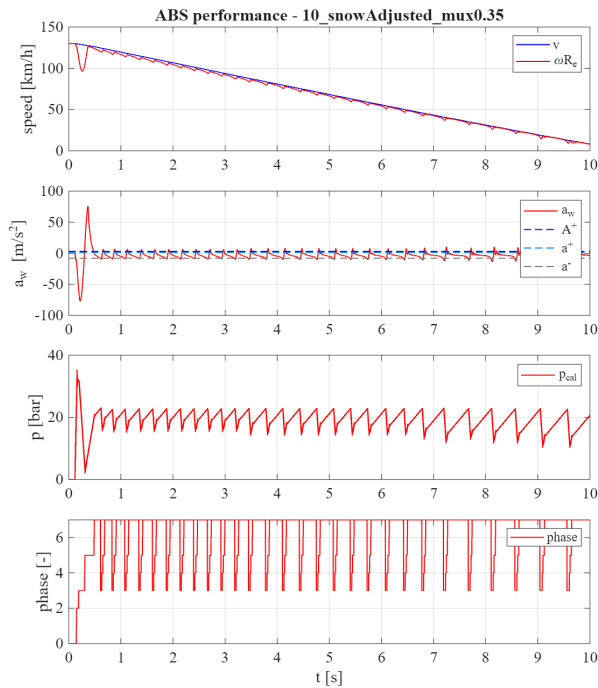
Using the robust 1.0 gains unchanged on snow. The thresholds (especially $a^- = -25$ m/s²) ask for far more deceleration than 0.35 grip can deliver, so the wheel spends a significant fraction of the braking event above the 15% slip threshold (nearly locked), and stopping distance balloons to 205 m.



- Snowy Adjusted Conditions

$\mu_x = -8$, $\mu_p = 1.5$, $\mu_a = 3$, $\mu_{dp5} = 250$, $\mu_{dp100} = 100$, $\mu_{dp7} = 20$ bar/s, $\mu_x = 0.35$

All three deceleration thresholds scaled down roughly in proportion to the grip drop ($a^-: -25 \rightarrow -8$, $a^+: 5 \rightarrow 1.5$, $A^+: 10 \rightarrow 3$) so the controller reacts to the much smaller decelerations achievable on snow instead of waiting for thresholds the tire can never reach; pressure rates also nearly halved ($500/200/40 \rightarrow 250/100/20$) for finer modulation at low grip. Shortens stopping distance from 205 m to 188 m and reduces slip overshoot from 16.9% to 14.0%.



- Snow Slip Comparison

From the two figures, it can be seen that the slip has been reduced after adjusting the ABS parameters for the lower grip.

